

Raphaël Bulle

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Nationality: French

Current position

Since October 2024, I am a **post-doctoral researcher** in the team-project MIMESIS at Centre Inria de l'Université de Lorraine in Strasbourg (funding: <https://anr.fr/Projet-ANR-22-CE46-0003>).

My current work focuses on the derivation of a **a posteriori error estimation** and **adaptive mesh refinement schemes** for the φ -FEM immersed boundary finite element method. I am also working on the use of φ -FEM to perform **shape optimization** via level-set functions. At MIMESIS my work aims at developing numerical tools for **bio-mechanics** and **biomedical simulation**.

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Research topics

Fields: Numerical analysis • Scientific calculus • Bio-mechanics

Keywords: Finite element methods • A posteriori error estimation • Immersed boundary methods • Fractional PDEs • Shape optimization

Professional experience

Research

10/2024 - Current	Post-doc A posteriori error estimation for the φ -FEM immersed boundary method applied to surgical stent manufacturing	MIMESIS, Centre Inria de l'U. de Lorraine
05/2022 - 08/2024	Post-doc Gradient recovery for multi-material finite element problems, collaboration with Michelin	GIREF, U. Laval, CA

Teaching

01/2024 - 05/2024	Lecture & exercises: Numerical analysis for the engineer Bachelor of engineering (100-110 students)	U. Laval, CA
09/2018 - 01/2019	Exercises: Mathematics for the engineer Bachelor of engineering 1st year (50-60 students)	U. Luxembourg
09/2015 - 01/2016	Lecture & exercises: Linear ODEs Bachelor of biology 1st year, 1/2 semester (25-35 students)	U. Franche-Comté
09/2014 - 01/2015	Exercises: Linear ODEs Bachelor of biology 1st year, 1/2 semester (25-35 students)	U. Franche-Comté

Formation

	PhD in Computational engineering and Mathematics Supervisors: S. P. A. Bordas, F. Chouly, J. S. Hale, A. Lozinski. Defended on the 25th of March 2022. Jury: Chair: Prof. O. Francis (U. Luxembourg)		
2022	Reviewers: • Prof. M. Vohralik (Inria Paris) • Prof. A. Bonito (Texas A&M, US) Supervisors: • Prof. S. P. A. Bordas (U. Luxembourg) • Prof. F. Chouly (U. de la República, Uruguay) • Prof. A. Lozinski (U. Franche-Comté)	Members: • Dr. J. Hale (U. Luxembourg) • Prof. R. Becker (U. Pau)	U. Luxembourg & U. Franche-Comté
2017	Master Advanced mathematics Master thesis: <i>A study of the Bank–Weiser a posteriori error estimator for finite element methods of arbitrary order</i> Supervision: Prof. F. Chouly (U. de la República, Uruguay), Prof. A. Lozinski (U. de Franche-Comté)		U. Franche-Comté
2016	Agrégation externe de Mathématiques	Education competitive exam.	National, FR
2015	Master Mathematics education		U. Franche-Comté
2014	Concours du CAPES de Mathématiques	Education competitive exam.	National, FR
2013	Bachelor Mathematics		U. Franche-Comté
2010	Baccalauréat Science, maths speciality		Lycée Louis Pergaud, Besançon

Supervision

Thesis

10/2025	Louis Ducongé.	MIMESIS, Centre
-	Topic: Shape optimization applied to surgical stents manufacturing.	Inria de l'U. de
Current	Co-supervisors: Dr. S. Cotin, Dr. M. Duprez, Pr. Y. Privat (U. de Lorraine)	Lorraine

Internship

03/2025	Louis Ducongé.	MIMESIS, Centre
-	Topic: Shape optimization applied to surgical stents manufacturing.	Inria de l'U. de
08/2025	Co-supervisors: Dr. S. Cotin, Dr. M. Duprez, Pr. Y. Privat (U. de Lorraine)	Lorraine
03/2024	Sarah Ait-Said.	
-	Topic: Gradient reconstruction for multi-material problems.	U. Laval, CA
08/2024	Co-supervisor: Prof. J. Deteix	

Scientific works

Publications

- 2026 *An adaptive multimesh rational approximation scheme for the spectral fractional Laplacian*
A. Bespalov, **R. B.**, Accepted in *SIAM Journal of Scientific Computing*, (Preprint: arxiv.org/abs/2504.03408)
- 2023 *An a posteriori error estimator for the spectral fractional power of the Laplacian*
R. B., O. Barrera, S.P.A. Bordas, F. Chouly, J.S. Hale, *Computer Methods in Applied Mechanics and Engineering*, doi.org/10.1016/j.cma.2023.115943
- 2023 *Hierarchical a posteriori error estimation of Bank-Weiser type in the FEniCS project*
R. B., J.S. Hale, A. Lozinski, S.P.A. Bordas, F. Chouly, *Computers & Mathematics with Applications*, doi.org/10.1016/j.camwa.2022.11.009
- 2021 *The human meniscus behaves as a functionally graded fractional porous medium*
R. B., G. Alotta, G. Marchiori, M. Berni, N. F. Lopomo, S. Zaffagnini, S. P. A. Bordas, O. Barrera, *Applied Sciences*, doi:10.3390/app11209405
- 2020 *Removing the saturation assumption in Bank-Weiser error estimator analysis in dimension three*
R. B., F. Chouly, J. S. Hale, A. Lozinski, *Applied Mathematics Letters*, doi:10.1016/j.aml.2020.106429

Submitted

- 2026 *Residual-based a posteriori error estimates with boundary correction for φ -FEM*
R. Becker, **R. B.**, M. Duprez, V. Lleras (Preprint: hal.science/hal-04931977v2)
- 2026 *A penalized ϕ -FEM scheme for the Poisson Dirichlet problem*
R. B., M. Duprez, V. Lleras, K. Vuillemot (Preprint: doi.org/10.48550/arXiv.2602.05698)

Ongoing work

- *Shape optimization using φ -FEM*
J. D. D. Avalos, **R. B.**, S. Cotin, L. Ducongé, M. Duprez, A. Laurain
- *Convergence of a rational scheme for the approximation of the spectral fractional Laplacian*
R. B.

Scientific organization

Chairman

- | | | |
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| 2025 | Contributed talks session | Biennial Num. An. Conf. |
| 2021 | Mini-symposium Advanced adaptive discretization methods | SIAM CSE |

Visits on invitation

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| Sep.
2023 | Invited by Dr. Alex Bespalov to work on a multimesh numerical scheme for the spectral fractional Laplacian.
Funded by the London Mathematical Society via Scheme 4 (Research in Pairs). | U. Birmingham,
GB |
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Research grants

NeuralFEM (MATH-AmSud) (22520 €)

Optimal and certified neural network predictions thanks to tools coming from finite element methods.

Coordinator: M. Duprez (Inria, France)

Collaborators:

2025

- J. P. Borthagaray, F. Chouly (U. de la República, Uruguay)
- M. Bastidas, D. A. Muñoz (U. Nacional de Colombia, Columbia)
- N. Guarín-Zapata (U. EAFIT, Columbia)
- R. Araya (U. de Concepción, e)
- P. Vega (U. de Santiago de Chile, Chile)
- J. Aguayo (U. del Bío–Bío, Chile)
- P. Alvarez, **R.B.**, S. Cotin (Inria, France)
- M. Genet (Ecole Polytechnique, CNRS, France)
- V. Ehrlacher, A. Ern (ENPC, CNRS, France)

Software

phiFEM (2025)

Convenience python package to use any of the φ -FEM schemes in the FEniCSx high performance finite element computing platform.

Repository: <https://github.com/PhiFEM/phiFEM>

Pypi: <https://pypi.org/project/phiFEM/>

DOI: <https://doi.org/10.6084/m9.figshare.30373546>

Language: Python

Development: ongoing

Developer: **R. B.**

Maintenance: yes

fenicsx-error-estimation (2021)

Implementation of the Bank–Weiser a posteriori error estimator in the FEniCSx high performance finite element computing platform.

Repository: <https://github.com/jhale/fenicsx-error-estimation>

DOI: https://figshare.com/articles/software/FEniCS_Error_Estimation_FEniCS-EE_/10732421

Languages: Python, C++

Development: paused

Developers: **R. B.**, J. S. Hale

Maintenance: no

Communications

International conferences on invitation

2026	<i>An adaptive multi-mesh rational approximation scheme for the spectral fractional Laplacian</i> (invited by Dr. A. Bespalov) R. B. , A. Bespalov	CMAM
2024	<i>Adaptive multi-mesh FEM for the spectral fractional Laplacian</i> (invited by Prof. S. Harizanov) R. B. , A. Bespalov	NMSCAA
2023	<i>An a posteriori error estimator for the spectral fractional power of the Laplacian</i> (invited by Prof. S. Harizanov) R. B. , O. Barrera, S. P. A. Bordas, F. Chouly, J. S. Hale	LSSC

International conferences and events

2025	<i>φ-FEM: a simple yet optimal immersed boundary finite element method</i> R. B.	SOFA Week
2025	<i>Residual a posteriori error estimates with boundary correction for φ-FEM</i> R. Becker, R. B. , M. Duprez, V. Lleras	Biennial Num. An. Conf.
2025	<i>Residual a posteriori error estimates with boundary correction for φ-FEM</i> R. Becker, R. B. , M. Duprez, V. Lleras	FEniCS conf.
2021	<i>Local a posteriori error estimates for the spectral fractional Laplacian</i> R. B. , S. P. A. Bordas, F. Chouly, J. S. Hale, A. Lozinski	FEniCS conf.
2021	<i>Practical aspects of a hierarchical a posteriori error estimator of Bank-Weiser type</i> R. B. , S. P. A. Bordas, J. S. Hale, F. Chouly, A. Lozinski	SIAM CSE
2021	<i>A posteriori error estimation for the fractional Laplacian</i> (poster) R. B. , A. Lozinski, F. Chouly, S. P. A. Bordas, J. S. Hale, doi:10.13140/RG.2.2.10144.00006	One Nonlocal World opening event
2020	<i>Practical aspects of the Bank-Weiser estimator implementation and bio-mechanics applications</i> R. B. , S. P. A. Bordas, J. S. Hale, F. Chouly, A. Lozinski	WCCM ECCOMAS Congress

Seminars

2026	<i>Adaptive multi-mesh FEM for the spectral fractional Laplacian</i> A. Bespalov, R. B.	Séminaire de l'IRMA U. de Strasbourg
2025	<i>Adaptive multi-mesh FEM for the spectral fractional Laplacian</i> A. Bespalov, R. B.	LEGATO team seminar U. Luxembourg
2025	<i>A posteriori error estimation and mesh adaptation</i> R. B.	Séminaire d'équipe MIMESIS Centre Inria de l'U de Lorraine
2025	<i>Adaptive multi-mesh FEM for the spectral fractional Laplacian</i> A. Bespalov, R. B.	Séminaire MACS IMAG, U. Montpellier
2024	<i>Estimation d'erreur a posteriori hiérarchique par reconstruction de gradients dans des milieux hétérogènes</i> R. B.	Séminaire Copilote Michelin-GIREF, U. Laval, CA
2022	<i>A posteriori error estimation in the FEniCSx finite element software and application to the fractional Laplacian</i> R. B. , S. P. A. Bordas, F. Chouly, J. S. Hale, A. Lozinski	Café technique Michelin, Clermont-Ferrand
2022	<i>Hierarchical a posteriori error estimation in the FEniCS finite element software and applications to fractional PDEs</i> R. B. , S. P. A. Bordas, J. S. Hale, F. Chouly, A. Lozinski	Séminaire SIMMUL, U. Laval, CA
2021	<i>Méthodes éléments finis et estimation d'erreur pour l'étude du ménisque</i> R. B. , S. P. A. Bordas, J. S. Hale, F. Chouly, A. Lozinski, O. Barrera	Mini-conférence PASS-SPI, U. Franche-Comté
2021	<i>Discretization of the fractional Laplacian using finite element methods and a posteriori error estimation</i> R. B. , S. P. A. Bordas, J. S. Hale, F. Chouly, A. Lozinski	Séminaire des doctorant-e-s U. Franche-Comté
2019	<i>Controlling error in multi-level approximations of stochastic PDEs</i> R. B. , F. Chouly, A. Lozinski, S.P.A. Bordas, J.S. Hale	Séminaire SPOC, IMB, U. de Bourgogne

Technical skills

- **Operating systems:** Linux, macOS, Windows.
- **Programming languages:** Python, C++, Matlab, Markdown, $\text{\LaTeX}$, bash.
- **Computation software:** FEniCSx, NetGen, MEF++, FreeFem++.
- **Other software:** Git, Docker/Podman, Conda/Mamba/Pixi, Quarto, Paraview, VS Code, Zotero, Vim.